Semantic Knowledge Graphs and Transformer Models for Accelerated Discovery of Photocatalytic Materials

Xinyi Cui

May 2025

Abstract

Photocatalytic materials are pivotal for sustainable hydrogen production via water splitting, yet their discovery is slowed by labor-intensive experiments and complex material-property relationships. This proposal presents a novel integration of semantic knowledge graphs (KGs) and graph-augmented transformer models to accelerate the identification of efficient photocatalysts. KGs integrate structured and unstructured data, enabling semantic reasoning to pinpoint promising candidates. Transformers predict critical properties like bandgap and hydrogen evolution rate (HER) with high accuracy. This hybrid approach is expected to reduce experimental iterations by 50% and discover novel photocatalysts with superior performance. Challenges include reconciling heterogeneous data sources and ensuring model interpretability. By streamlining photocatalyst design, this work advances sustainable energy solutions.

1 Introduction

Photocatalytic water splitting offers a sustainable pathway for hydrogen production, addressing global energy and environmental challenges [1]. Materials like TiO_2, $BiVO_4$, and g-C_3N_4 show promise, but their discovery is slowed by trial-and-error experimentation and the complexity of optimizing properties like light absorption and charge separation [2]. Artificial Intelligence (AI) has revolutionized materials science. Machine learning (ML) models, such as Random Forests and Convolutional Neural Networks (CNNs), predict bandgap with root mean square errors (RMSE) below 0.4 eV [1], [3]. High-throughput computational workflows have screened millions of compounds [4], and semantic knowledge graphs (KGs) have integrated heterogeneous data for enhanced reasoning [5], [6]. Using Connected Papers, we identified [1] as a foundational study on ML-driven photocatalyst discovery, extended by [5]'s KG approach for data integration and [2]'s survey of transformer-based methods.

Despite these advances, challenges remain. ML models often lack interpretability, and data heterogeneity complicates integration [5]. Existing approaches struggle to navigate the vast chemical space of photocatalysts, limiting novel heterojunction discovery. This proposal combines KGs for data integration and transformer models for property prediction to accelerate the identification of high-performance photocatalysts.

2 Problem Statement & AI Approach

Research Question: How can semantic knowledge graphs and transformer models predict and optimize the photocatalytic performance of novel materials for hydrogen production?

The challenge is identifying photocatalysts with optimal bandgap (1.23–3 eV), high charge separation efficiency, and stability under illumination. Traditional density functional theory (DFT) simulations and experimental screening are time-intensive [1]. We propose a hybrid AI approach:

1. **Semantic Knowledge Graphs**: KGs will integrate structured (e.g., Materials Project) and unstructured (e.g., scholarly articles) data. Nodes represent materials (e.g., TiO_2), properties (e.g., bandgap = 3.2 eV), and conditions (e.g., pH), with edges like "*has_property*" or "*is_similar_to*". Semantic reasoning identifies candidates with desirable traits [5], [7]. The KG's human-readable relationships enhance interpretability of subsequent predictions.

2. **Transformer Models**: A graph-augmented transformer, fine-tuned on KG-derived node embeddings and molecular descriptors, will predict bandgap and hydrogen evolution rate (HER). This leverages transformers' ability to model non-linear relationships, improving accuracy over traditional ML [2], [8]. Transformers were chosen over generative models like GANs due to their superior handling of graph-structured data and non-linear property relationships, critical for complex chemical spaces.

This approach outperforms traditional ML by leveraging transformers' modeling power and KGs' contextual insights, with the KG narrowing the chemical space and transformers predicting properties for top candidates. The KG narrows the chemical space, and the transformer predicts properties for top candidates, forming an iterative pipeline inspired by [5].

3 Data Considerations & AI Workflow

Data Sources: The Open Catalyst Project provides DFT-computed properties, and the Materials Project offers crystallographic data. Unstructured data from scholarly articles will be extracted using SciBERT-based information extraction [5].

Data Challenges: Heterogeneity requires robust preprocessing. Biases (e.g., overrepresentation of TiO_2) may skew predictions. Favoring well-studied materials could limit exploration of underrepresented compounds, raising ethical concerns about equitable research focus. We will use synthetic data generation via SMILES-based generative models and oversampling underrepresented materials to mitigate biases. Named Entity Recognition (NER) with spaCy and ontology matching will standardize entities [6].

Materials Toolkits: PymatGen will calculate descriptors (e.g., electronegativity), Matminer will preprocess data, and DeepChem will extract molecular features [4].

Workflow:

1. *KG Construction*: Use NER and relation extraction to build a KG with nodes (materials, properties) and edges (relationships).

2. *Feature Extraction*: Generate node embeddings using Graph Neural Networks (GNNs) and molecular descriptors.

3. *Transformer Training*: Train a graph-augmented transformer on embeddings to predict bandgap and HER. Monitor RMSE for bandgap and Pearson correlation for HER during training to ensure convergence.

4. *Candidate Ranking*: Rank materials by predicted properties and novelty (Tanimoto similarity < 0.8).

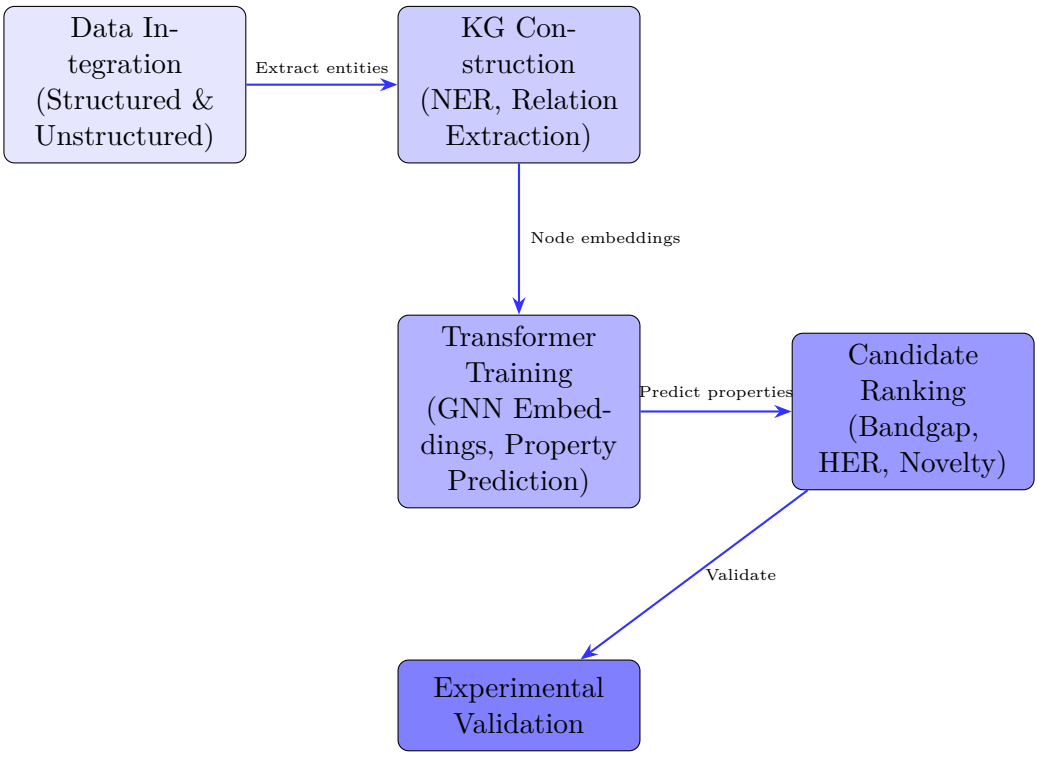


Figure 1: Proposed AI workflow: Semantic knowledge graph integrates heterogeneous data, transformer predicts properties using GNN embeddings, and candidates are ranked for experimental validation.

4 Expected Outcomes & Evaluation Metrics

Predicted Impact: The approach will reduce experimental iterations by 50%, as demonstrated in similar AI-driven workflows [4], and identify photocatalysts with HER > 6 mmol g^{-1} h^{-1}, surpassing benchmarks [1]. By mapping semantic relationships, it may reveal novel structure-property correlations, advancing photocatalyst design principles. It prioritizes earth-abundant materials for sustainability.

Success Criteria:

Table 1: Evaluation Metrics and Targets

Metric	Target	Benchmark
Bandgap RMSE	< 0.3 eV	DFT simulations
HER Pearson Correlation	> 0.8	Experimental HER
Novelty (Tanimoto Similarity)	< 0.8	Materials Project

Benchmarking: Predictions will be validated against DFT simulations and experimental HER measurements. Discrepancies will be addressed using uncertainty quantification (e.g., ensemble models). High-throughput screening results serve as baselines [4].

5 Challenges & Limitations

Challenges: Transformer training and KG construction require $\sim$ 200 GPU hours on an NVIDIA A100 for 10,000 compounds. Transformers are black-box models, limiting interpretability [6]. Data biases may reduce generalizability. A failure case could be overfitting to biased training data, leading to poor generalization to novel material classes.

Mitigation: Use transfer learning and cloud computing (e.g., AWS) to reduce costs. Apply SHAP for feature importance and attention visualization for interpretability. Mitigate biases with synthetic data and regular audits. Validate against diverse datasets to ensure generalizability.

Scalability: The approach is adaptable to other domains (e.g., battery electrolytes) by modifying the KG ontology and retraining the transformer, leveraging tools like PyTorch and Deep Graph Library (DGL).

6 Project Timeline

Table 2: Project Roadmap

Phase	Duration
KG Construction	6 months
Transformer Training	4 months
Prediction and Ranking	2 months
Experimental Validation	3 months

The 15-month timeline ensures thorough development and validation, with milestones for data integration, model training, and experimental confirmation.

7 Conclusion

This proposal integrates semantic KGs and transformer models to accelerate photocatalytic material discovery, addressing data heterogeneity and prediction challenges. The approach promises novel photocatalysts with HER > 6 mmol g^{-1} h^{-1}, advancing sustainable hydrogen production. Next steps include constructing the KG using SciBERT, training a graph-augmented transformer, and validating predictions experimentally. Future refinements will enhance interpretability through attention-based visualizations to guide experimental design. Future work will enhance interpretability and extend to other material classes.

References

[1] H. Masood, C. Y. Toe, W. Y. Teoh, V. Sethu, and R. Amal, "Machine learning for accelerated discovery of solar photocatalysts," *ACS Catalysis*, vol. 9, no. 12, pp. 11 774–11 787, Dec. 2019. DOI: 10.1021/acscatal.9b02531.

[2] Y. Xu, H. Wang, W. Zhang, *et al.*, "Ai-empowered catalyst discovery: A survey from classical machine learning approaches to large language models," *arXiv*, Feb. 2025. DOI: 10.48550/arXiv.2502.13626.

[3] T. Xie, Y. Wan, H. Wang, *et al.*, "Opinion mining by convolutional neural networks for maximizing discoverability of nanomaterials," *Journal of Chemical Information and Modeling*, vol. 64, no. 7, pp. 2746–2759, Apr. 2024. DOI: 10.1021/acs.jcim.3c00746.

[4] F. Delgado-Licona and M. Abolhasani, "Research acceleration in self-driving labs: Technological roadmap toward accelerated materials and molecular discovery," *Advanced Intelligent Systems*, vol. 5, no. 4, p. 2 200 331, Apr. 2023. DOI: 10.1002/aisy.202200331.

[5] V. Venugopal and E. Olivetti, "Matkg: An autonomously generated knowledge graph in material science," *Scientific Data*, vol. 11, no. 1, p. 217, Feb. 2024. DOI: 10.1038/s41597-024-03039-z.

[6] V. Venugopal, S. Pai, and E. Olivetti, "The largest knowledge graph in materials science—entities, relations, and link prediction through graph representation learning," in *AI for Accelerated Materials Design NeurIPS 2022 Workshop*, Nov. 2022. [Online]. Available: https://openreview.net/forum?id=xyJ_0-WCIZN.

[7] P. Guo, W. Meng, and Y. Bao, "Knowledge graph-guided data-driven design of ultra-high-performance concrete (uhpc) with interpretability and physicochemical reaction discovery capability," *Construction and Building Materials*, vol. 430, p. 136 502, Jun. 2024. DOI: 10.1016/j.conbuildmat.2024.136502.

[8] J. Liu, L. Liang, B. Su, *et al.*, "Transformative strategies in photocatalyst design: Merging computational methods and deep learning," *Journal of Materials Informatics*, vol. 4, no. 4, N/A, 2024. DOI: 10.20517/jmi.2024.48.